Quadratic Coefficients

Two Variables Example

(x1 x2) [a11 a12] [x1] = (a11 x1 + a21 x2 a12 x1 + a22 x2) [x1]
[a21 a22] [x2] (a11 x1 + a21 x2)x1 + (a12 x1 + a22 x2)x2
= a11 x1^2 + a21 x1 x2 + a12 x1 x2 + a22 x2^2
= a11 x1^2 + (a21 + a12) x1 x2 + a22 x2^2 => Quadratic form

a11 => Coefficient of x1^2
a21 + a12 => Coefficient of x1 x2 => a12 = a21 = 1/2 Coefficient of x1 x2
a22 => Coefficient of x2^2

(x1 x2) [a11 a12] => [coef of x1^2 0.5 coef of x1 x2]
[a21 a22] [0.5 coef of x1 x2 coef of x2^2]
(Note: symmetric matrix - off-diagonal elements are half the cross-term coefficient)